

The empirical recurrence for OEIS A232080: recovering a conjecture that is not written down

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Abstract

OEIS A232080 records “Empirical recurrence of order 54”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 78$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 54, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 54 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A232080 is “Number of $(4+1)X(n+1)$ 0..1 arrays with every element equal to some horizontal, diagonal or antidiagonal neighbor, with top left element zero.”. It has offset 1 and begins

111, 3985, 82996, 2265691, 64298979, 1786680190, 49535329657, 1375366780223, ...

The entry states:

Empirical recurrence of order 54 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 06:52:28, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 1056$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 78$ of the 1056, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 78$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 2112$ exact terms of the model returns order 54 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{54} c_i a(n-i),$$

c_1	17	c_{15}	−5466020520	c_{29}	1372177045551	c_{43}	75680710988
c_2	199	c_{16}	−7174706705	c_{30}	1690947992899	c_{44}	12159359476
c_3	2095	c_{17}	−23109502971	c_{31}	−3333605114781	c_{45}	−17846423736
c_4	17163	c_{18}	10347448009	c_{32}	−68242896950	c_{46}	7876933224
c_5	49164	c_{19}	92285895608	c_{33}	−1400539218595	c_{47}	−1393610496
c_6	5728	c_{20}	52003875308	c_{34}	−1487594626433	c_{48}	−1790080416
c_7	−982225	c_{21}	189224703657	c_{35}	2268960028309	c_{49}	1186644672
c_8	−9162832	c_{22}	31476357249	c_{36}	−181589577357	c_{50}	−105946176
c_9	−22727679	c_{23}	−674091223527	c_{37}	611518034246	c_{51}	−44017920
c_{10}	22600243	c_{24}	−215277659729	c_{38}	462658501938	c_{52}	51867648
c_{11}	137134660	c_{25}	−715597658065	c_{39}	−623586266026	c_{53}	−11612160
c_{12}	472206655	c_{26}	−562065153852	c_{40}	−461025094	c_{54}	−2239488
c_{13}	1184571797	c_{27}	2168538834706	c_{41}	−16432639110		
c_{14}	−1171631903	c_{28}	323794079026	c_{42}	−89621042484		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 54, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $54 + 4$ terms removed returns order 54 as well, so $\deg q_{\min} = 54 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 15 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $54 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 54 that this sequence satisfies — and that family turns out to have exactly one member.

References

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